Optimizing Egg Drop Strategies: Correcting the Asymptotic Constants for the 1-3 and 2-2 Approaches

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Abstract

In a recent paper on higher-dimensional egg drop problems, the bound $P_1(k) \leq kN^{1/k}$ was used to compare two strategies for four eggs: the **1-3** approach (one coarse egg, then three eggs) and the **2-2** approach (two eggs on superfloors, then two eggs inside a block). That bound is **not tight**. The exact minimax number of drops is determined by the partial sums of binomial coefficients: with d drops and k eggs one can test at most $\sum_{j=1}^{k} \binom{d}{j}$ floors. Using this exact formula, we re-optimize both strategies for large N. We find that the 1-3 strategy achieves a leading constant of approximately **2.746**, while the 2-2 strategy achieves **2.828**. Thus, contrary to the earlier conclusion, the **1-3 strategy is asymptotically better** for four eggs and a large number of floors. We provide all derivations and a downloadable PDF.

Keywords: egg drop, minimax search, binomial coefficients, asymptotic constants, 1-3 strategy, 2-2 strategy

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1. Introduction

The classical egg drop problem asks for the fewest number of drops needed to identify the critical floor in an N-story building with k eggs. The paper *Egg Drop Problems: They Are All They Are Cracked Up to Be!* proved the upper bound

$$P_1(k) \leq \lceil kN^{1/k} \rceil.$$

This bound is convenient but **not tight**. The exact minimax capacity is known from the work of Boardman (2004) and others:

> With d drops and k eggs, the maximum number of floors that can be tested is
>
> $$E(d,k) = \sum_{j=1}^{k} \binom{d}{j}.$$

Equivalently, the minimum number of drops required to cover N floors with k eggs is the smallest integer d such that

$$\sum_{j=1}^{k} \binom{d}{j} \geq N.$$

For large N, the leading term of this sum is $d^k/k!$, so

$$d \sim (k!\, N)^{1/k}.$$

Thus the correct asymptotic constant is $(k!)^{1/k}$, not k. For example:

k	Paper's bound $kN^{1/k}$	Exact leading term $(k!\, N)^{1/k}$
2	$2N^{1/2}$	$(2N)^{1/2} \approx 1.414N^{1/2}$
3	$3N^{1/3}$	$(6N)^{1/3} \approx 1.817N^{1/3}$
4	$4N^{1/4}$	$(24N)^{1/4} \approx 2.213N^{1/4}$

The paper's bound overestimates the true cost, which changes the comparison between the two strategies for four eggs.

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2. The Two Strategies

We consider **four eggs** and N floors.

2.1 The 1-3 Strategy

1. Use **one egg** to make coarse jumps of size S. Drop it at floors $S, 2S, 3S, \ldots$ until it breaks.
2. When it breaks, the critical floor lies in an interval of length at most S.
3. Use the remaining **three eggs** to search that interval optimally.

Worst-case drops:

- Coarse jumps: at most N/S.
- Three-egg search on an interval of size S: the exact number is the smallest d with $\sum_{j=1}^{3} \binom{d}{j} \geq S$.

For large S, the three-egg search requires approximately

$$(6S)^{1/3}$$

drops. Hence the total is roughly

$$D_{\text{1-3}}(S) \approx \frac{N}{S} + (6S)^{1/3}.$$

2.2 The 2-2 Strategy

1. Partition the N floors into blocks of size F. There are N/F superfloors.
2. Use **two eggs** to find the critical superfloor (the block containing the breaking point). This is a standard two-egg search on N/F items.
3. Use the remaining **two eggs** to search inside the critical block of F floors.

Worst-case drops:

- Two-egg search on N/F superfloors: the exact number is the smallest d with $\binom{d}{1} + \binom{d}{2} \geq N/F$, which for large N/F is approximately

$$\sqrt{2\,(N/F)}.$$

- Two-egg search inside the block of size F: approximately

$$\sqrt{2F}.$$

Thus

$$D_{2\text{-}2}(F) \approx \sqrt{2}\,\frac{N^{1/2}}{F^{1/2}} + \sqrt{2}\,F^{1/2}.$$

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3. Optimizing the 1-3 Strategy

We minimize

$$f(S) = \frac{N}{S} + (6S)^{1/3}, \qquad S > 0.$$

Differentiate:

$$f'(S) = -\frac{N}{S^2} + \frac{1}{3}(6S)^{-2/3} \cdot 6 = -\frac{N}{S^2} + 2\,(6S)^{-2/3}.$$

Set $f'(S) = 0$:

$$\frac{N}{S^2} = 2\,(6S)^{-2/3} \implies N = 2\,6^{-2/3} S^{4/3} \implies S^{4/3} = \frac{N\,6^{2/3}}{2}.$$

Thus the optimal step size scales as

$$S \propto N^{3/4}.$$

Write $S = cN^{3/4}$. Then

$$\frac{N}{S} = \frac{N^{1/4}}{c}, \qquad (6S)^{1/3} = (6c)^{1/3} N^{1/4}.$$

So

$$f = N^{1/4}\left(\frac{1}{c} + (6c)^{1/3}\right).$$

Minimize the bracket over $c > 0$:

$$g(c) = \frac{1}{c} + (6c)^{1/3}.$$

$$g'(c) = -\frac{1}{c^2} + \frac{1}{3}(6c)^{-2/3}\cdot 6 = -\frac{1}{c^2} + 2\,(6c)^{-2/3}.$$

Set $g'(c) = 0$:

$$\frac{1}{c^2} = 2\,(6c)^{-2/3} \implies c^{4/3} = \frac{1}{2\,6^{-2/3}} = \frac{6^{2/3}}{2}.$$

Thus

$$c = \left(\frac{6^{2/3}}{2}\right)^{3/4} = \frac{6^{1/2}}{2^{3/4}}.$$

Now evaluate the constant:

$$\frac{1}{c} = \frac{2^{3/4}}{6^{1/2}}, \qquad (6c)^{1/3} = 6^{1/3}c^{1/3} = 6^{1/3}\left(\frac{6^{1/2}}{2^{3/4}}\right)^{1/3} = \frac{6^{1/2}}{2^{1/4}}.$$

So

$$g_{\min} = \frac{2^{3/4}}{6^{1/2}} + \frac{6^{1/2}}{2^{1/4}}.$$

Numerically:

$$\frac{2^{3/4}}{6^{1/2}} \approx \frac{1.6818}{2.4495} \approx 0.6866, \qquad \frac{6^{1/2}}{2^{1/4}} \approx \frac{2.4495}{1.1892} \approx 2.0598.$$

Hence

$$g_{\min} \approx 2.7464.$$

Therefore

$$\boxed{D_{1\text{-}3} \sim 2.746\,N^{1/4}}.$$

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4. Optimizing the 2-2 Strategy

We minimize

$$h(F) = \sqrt{2}\,\frac{N^{1/2}}{F^{1/2}} + \sqrt{2}\,F^{1/2}, \qquad F > 0.$$

Factor out $\sqrt{2}$:

$$h(F) = \sqrt{2}\left(N^{1/2}F^{-1/2} + F^{1/2}\right).$$

Let

$$\phi(F) = N^{1/2}F^{-1/2} + F^{1/2}.$$

$$\phi'(F) = -\frac{1}{2}N^{1/2}F^{-3/2} + \frac{1}{2}F^{-1/2}.$$

Set $\phi'(F) = 0$:

$$N^{1/2}F^{-3/2} = F^{-1/2} \implies N^{1/2} = F.$$

Thus the optimal block size is

$$F = N^{1/2}.$$

Substitute back:

$$\phi_{\min} = N^{1/2}(N^{1/2})^{-1/2} + (N^{1/2})^{1/2} = N^{1/4} + N^{1/4} = 2N^{1/4}.$$

Therefore

$$h_{\min} = \sqrt{2} \cdot 2N^{1/4} = 2\sqrt{2}\, N^{1/4}.$$

Numerically:

$$2\sqrt{2} \approx 2.8284.$$

So

$$\boxed{D_{2\text{-}2} \sim 2.828\, N^{1/4}}.$$

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5. Comparison and Conclusion

Strategy	Leading constant
1-3	2.746
2-2	2.828

For large N, the **1-3 strategy** requires fewer drops in the worst case. The difference is about 2.9%.

This contradicts the earlier conclusion based on the non-tight bound $kN^{1/k}$, which incorrectly suggested that both strategies are asymptotically equivalent

(both giving constant 4).

The correct asymptotic constants follow from the exact binomial-sum capacity $\sum_{j=1}^{k} \binom{d}{j}$. The 1-3 strategy is slightly superior because it leverages the more efficient three-egg search in the second phase, whereas the 2-2 strategy uses the less efficient two-egg search twice.

For finite N, the optimal integer choices of S and F may shift the comparison, but the asymptotic advantage of the 1-3 strategy is clear.

Appendix: Exact Integer Optimization

For practical implementation, one should compute the exact minimum drops for a given N by evaluating

$$D_{\text{1-3}}(S) = \left\lceil \frac{N}{S} \right\rceil + d_3(S),$$

where $d_3(S) = \min\{d : \sum_{j=1}^{3} \binom{d}{j} \geq S\}$, and similarly for the 2-2 strategy:

$$D_{\text{2-2}}(F) = d_2\left(\frac{N}{F}\right) + d_2(F),$$

with $d_2(M) = \min\{d : \binom{d}{1} + \binom{d}{2} \geq M\}$.

These exact values can be computed efficiently and will give the precise best choice for any specific N.

References

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